

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                | Marks Available |             |     |       |       |      |
|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2         | AO3 | Total | Maths | Prac |
| 3        | (a) |  |  | Potassium } (1) carbonate (1)<br>Potassium } chloride (1)<br><br>Reference to CO <sub>2</sub> for sample 1 neutral                                                                                                                                                                                                                             |                 | 3           |     | 3     |       | 3    |
|          | (b) |  |  | 16 × 4 = 64 (1)<br>Addition: M <sub>r</sub> = 136 (1)<br>34/136 (1)<br>Number of moles = 0.25 (1)<br><br>For 3 Marks:<br>88 (1)<br>34 / 88 (1)<br>0.386 (1)<br><br>Use of 64 as M <sub>r</sub> to give 0.53 (3)<br><br>Use of 352 (88 × 4) as M <sub>r</sub> to give 0.097/ 0.09659 (3)<br><br>For 2 marks<br>34/candidate mass (1)<br>Ans (1) | 1               | 1<br>1<br>1 |     | 4     | 4     |      |
|          |     |  |  | Question 3 total                                                                                                                                                                                                                                                                                                                               | 1               | 6           |     | 7     | 4     | 3    |